



YISHUN INNOVA JUNIOR COLLEGE  
JC 2 MID YEAR EXAMINATION  
**Higher 2**

CANDIDATE  
NAME

CG

INDEX NO

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**COMPUTING**

Paper 1 Written

**9569/01**

**1 Jul 2022**

**3 hours**

Additional Materials: Nil

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**READ THESE INSTRUCTIONS FIRST**

Write your name, index number and class clearly on the cover page.

Write in dark blue or black pen on the writing paper provided.

You may use an HB pencil for any diagrams, graphs, tables or rough working.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

Approved calculators are allowed.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is **100**

- 1
- (a) The ASCII code in denary for the character 'Y' is 89.
- (i) Using 7 bits, express the ASCII code in binary for the character 'Y'. [2]
- (ii) Express the ASCII code for character 'J' as a hexadecimal number. [2]
- (b) ASCII uses 7 bits to represent 128 characters to be encoded. State the number of characters represented by the extended ASCII if it uses 8 bits. [1]
- (c) Given that the Unicode character 'µ' is represented as 03BC in the hexadecimal number system, convert it into a 2-byte binary number. [2]
- (d) Give **one** advantage and **one** disadvantage of using ASCII over Unicode. [2]
- (e) Describe how a 2-byte Unicode character can be transmitted over the internet. [2]
- 2
- An insurance company assigns a unique seven-character ID to each customer. The seventh character is a checksum character which is obtained from the following series of calculations:
- multiply each of the six digits by its position (position 1 is the leftmost position)
  - find the sum of the resulting six products
  - calculate the result of the sum modulo 11
  - the checksum character is 'X' if the result is 10; Otherwise, the checksum character is the value of the result.
- (a) Showing your working clearly for the calculation, find the checksum character for each of the following:
- (i) 632123 [2]
- (ii) 821321 [2]
- (b) Deduce whether 342341X is a valid ID. Show your working. [3]
- (c) Besides using the checksum character, state another two types of methods to validate a customer's ID. [2]

- 3 Customer IDs are stored in a Binary Search Tree (BST) and the Postal Code is used as the comparison key. If the Postal Codes are the same, the Unit Numbers will be used for comparison.

The Customer IDs are inserted into the BST in the following order:

| Customer ID | Unit Number | Postal Code |
|-------------|-------------|-------------|
| C01         | #06-01      | 524287      |
| C02         | #02-03      | 131071      |
| C03         | #01-01      | 768675      |
| C04         | #02-55      | 262884      |
| C05         | #01-01      | 104857      |
| C06         | #04-10      | 768675      |
| C07         | #15-03      | 131071      |

- (a) Draw the populated BST with the Customer ID as the data. [3]

The BST is implemented using an array of nodes and each node comprises the Customer ID, LPtr and RPtr.

- (b) Copy and complete the content of the array below:

| Index | Customer ID | LPtr | RPtr |
|-------|-------------|------|------|
| 0     | C01         |      |      |
| 1     |             |      |      |
| 2     |             |      |      |
| 3     |             |      |      |
| 4     |             |      |      |
| 5     |             |      |      |
| 6     |             |      |      |

[5]

- (c) Describe how you would:
- (i) obtain the outputs of the Customer IDs in the BST by a post-order traversal. [4]
  - (ii) search the BST for the node containing the Postal Code 262884 and the Unit Number #02-55. [2]
  - (iii) search the BST for the node containing the Postal Code 615100 and the Unit Number #03-10. [2]

- 4 An old school library in rural Cambodia uses a flat file to record the books borrowed by the students.

The librarian records the loan details in the following table:

| LoanID | BookID | Title                                | Author         | StudentID | StudentName | LoanDate       | ReturnDate  |
|--------|--------|--------------------------------------|----------------|-----------|-------------|----------------|-------------|
| 01     | 2132   | Around<br>The<br>World in<br>80 Days | Jules<br>Verne | 101       | Chea        | 2 Mar<br>2022  | 12 Mar 2022 |
| 02     | 1078   | Khmer<br>History                     | Tom<br>Smith   | 101       | Chea        | 2 Mar<br>2022  | 5 Mar 2022  |
| 03     | 2132   | Around<br>The<br>World in<br>80 Days | Jules<br>Verne | 102       | Sokha       | 13 Mar<br>2022 |             |

The `ReturnDate` field will be filled when the student returns the book.

- (a) Explain with example(s) why the table is not in second normal form (2NF). [2]
- (b) State the requirements for a table to be in third normal form (3NF). [2]

The library would like to design a relational database to store the data.

- (c) Construct three 3NF tables, **Book**, **Student**, and **Loan**, using the existing fields and data in the table provided. [3]

- (d) A table description can be expressed as:

TableName (Attribute1, Attribute2, Attribute3, ...)

The primary key is indicated by underlining one or more attributes. Foreign keys are indicated using a dashed underline.

Write table descriptions for the three tables in **part (c)**. [3]

- (e) Draw an Entity-Relationship (E-R) diagram showing the three tables and the relationships between them. [3]

The library would like to use the same database to store the loan data of an existing collection of music CDs. Typically, a music CD has an album title, name of the artist and the genre.

In future, the library intends to acquire some English and Khmer language audio books for students to borrow.

- (f) State **two** reasons why a NoSQL database might be more suitable for the library. [2]

5. (a) Explain, with example(s), the following terms with regard to user interface (UI) in web design:
- (i) Affordance [2]
  - (ii) Mental model. [2]

The following web page shows the school's library system which allows the students to log in using their email address and password to borrow books from the library.

## High School Loan System

Login

Name:

Email:

Password:

Book Selection:

You can borrow up to 4 items.  
Which book do you want to borrow?

Please choose from the 400 items in the dropdown.

Loan date (Please enter today's date):

Add in the titles of the additional items which you want to borrow:

[Submit](#)

Please note that for some items, the loan period is 1 week instead of 2.  
Failure to return on time will lead to disciplinary actions

- (b) Identify and describe **three** poor UI designs in the above web page. Suggest an improvement for each of the design issues identified and explain how they could improve the user experience. [6]

**6** A hospital building has a local area network (LAN) equipped with servers and workstations for the staff to use. The public can access the server via the internet to register or check their appointments.

- (a) State two advantages and one disadvantage of a Server-Client LAN. [3]
- (b) Describe how data is transmitted by packet switching. Your description should include the situation where some of the packets are lost during the transmission. [5]
- (c) Describe how the Transmission Control Protocol (TCP) and Internet Protocol (IP) are used to enable the communication between two computers over the internet. [3]
- (d) Describe how the Domain Name System (DNS) works in helping to retrieve the web page from the Amazon's web server when the user types `www.amazon.com` into the browser's address bar. [3]

Alice receives an email from the hospital's billing department, with the instructions to pay for her recent treatment at the hospital.

- (e) Explain how Digital Certificate and Digital Signature help to verify the authenticity and integrity of the email sent by the sender. [7]

- 7 The School Admin Portal has a form for the students to reset their accounts' login password. To reset the password, the student keys in his `UserID` and a new password.

**School Admin Portal**

Reset Password

**UserID :**

**New Password :**

**Confirm New Password :**

**Submit**

- (a) State the purpose of validation and verification in data entry. [2]
- (b) Describe how validation and verification are enforced in the above form. [2]
- (c) Explain why `HTTP POST` method should be used instead of the `HTTP GET` method for submitting the form to the back-end server. [2]
- (d) Describe how passwords are stored in the database and how they are being used for authentication when a user logs in with his `UserID` and password. [2]



- 8      (a)      (i)      Define *personal data*. [2]
- (ii)      State any **two** of the obligations under the Personal Data Protection Act (PDPA). [2]

Bob is the enrolment administrator at the Star University.

During the admissions exercise, he sent a mass email to remind the applicants to submit the completed personal particulars form. The email contained the names and email addresses of all the applicants.

When some applicants approached Bob for help to complete the personal particulars form, he showed them another applicant's completed form as a sample for their reference.

- (b)      State and explain how Bob's actions have violated the obligations in the PDPA. [4]
- (c)      For each of the violations stated in **part (b)**, discuss the correct actions that Bob should have taken to protect the personal data of the applicants. [2]

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